

component.

- 3 **A** By Newton's second law, taking the direction of F_1 as positive,

$$F_{net} = ma$$

$$F_1 - F_2 = ma$$

$$a = \frac{F_1 - F_2}{m}$$

Since object was at rest, F_2 is initially zero ($\because F_2 \propto v^2$) and acceleration is maximum initially. Acceleration then decreases as F_2 increases with increasing speed as object accelerates. Acceleration becomes zero eventually when $F_1 = F_2$ and object travels at constant (maximum) speed henceforth.

- 4 **A** At equilibrium, the lines of action of three forces must meet at a point